

Haldane Quantum Dot - Benchmark Report

Generated: 2025-10-22 01:04:35

Run configuration:

t: 1.0

t2: 0.1

phi: 1.5707963267948966

M: 0.2

R: 15.0

edge_frac: 0.2

k_eigs: 120

seed: 0

shape: disk

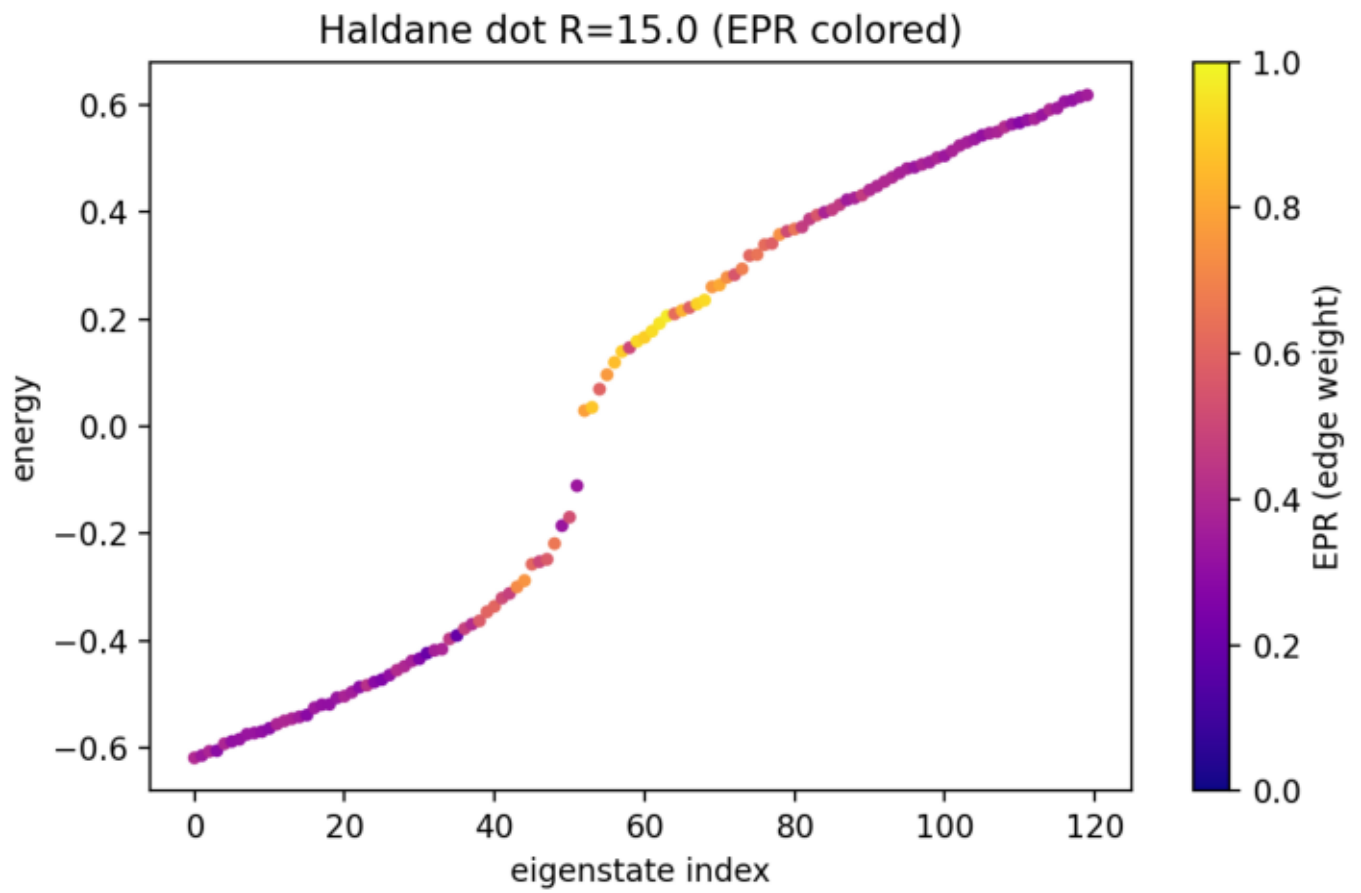
Single-dot metrics

hermiticity_error: 0.0
min_abs_energy: 0.029304431757409644
num_sites: 1630
seed: 0
mean_rim_current: 0.00023735464198524934
shape: disk

Size-sweep fit metrics

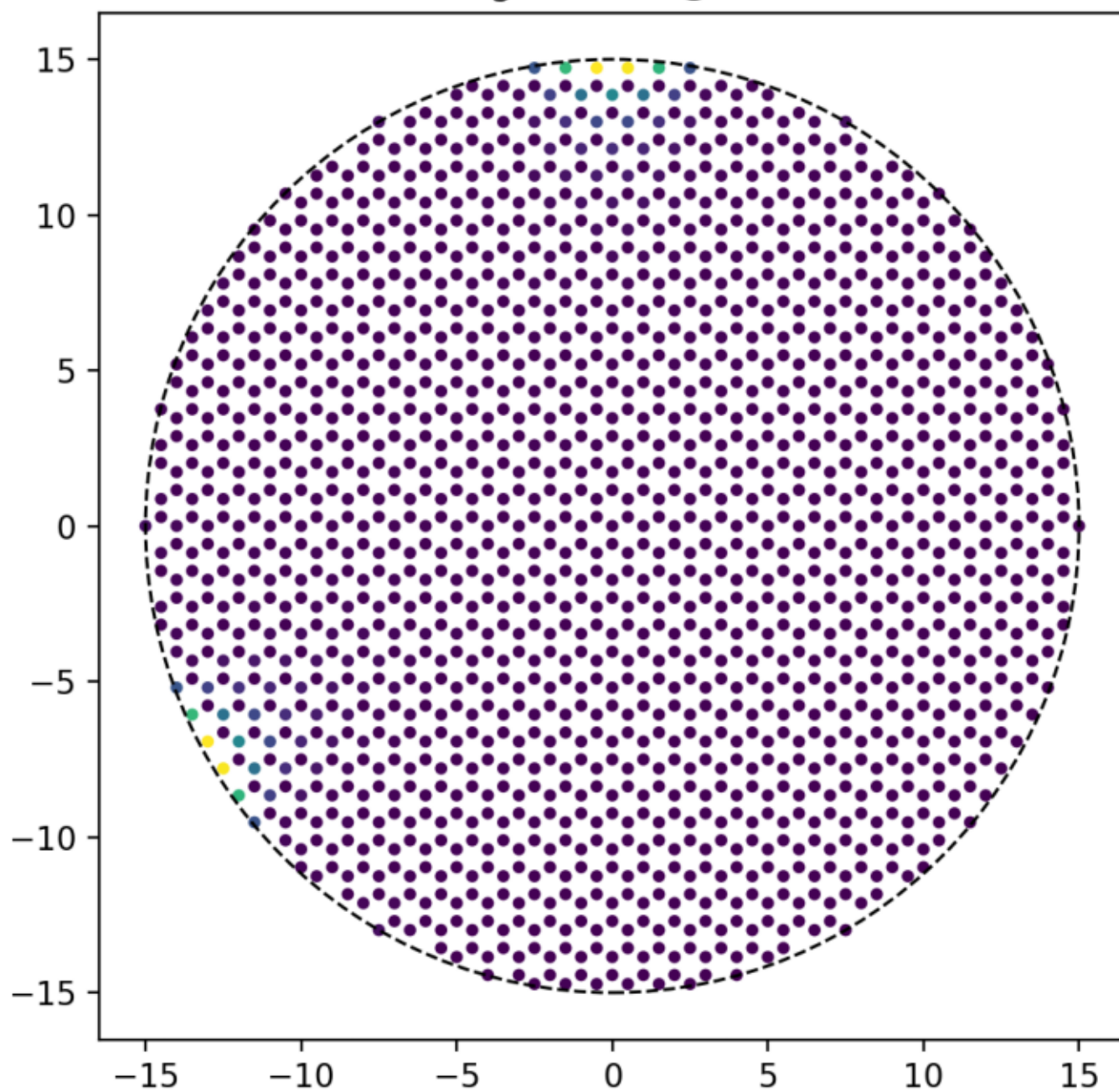
v_{edge}: 0.268292692021396
r²: 0.9488905012969939

Spectrum (EPR-colored)

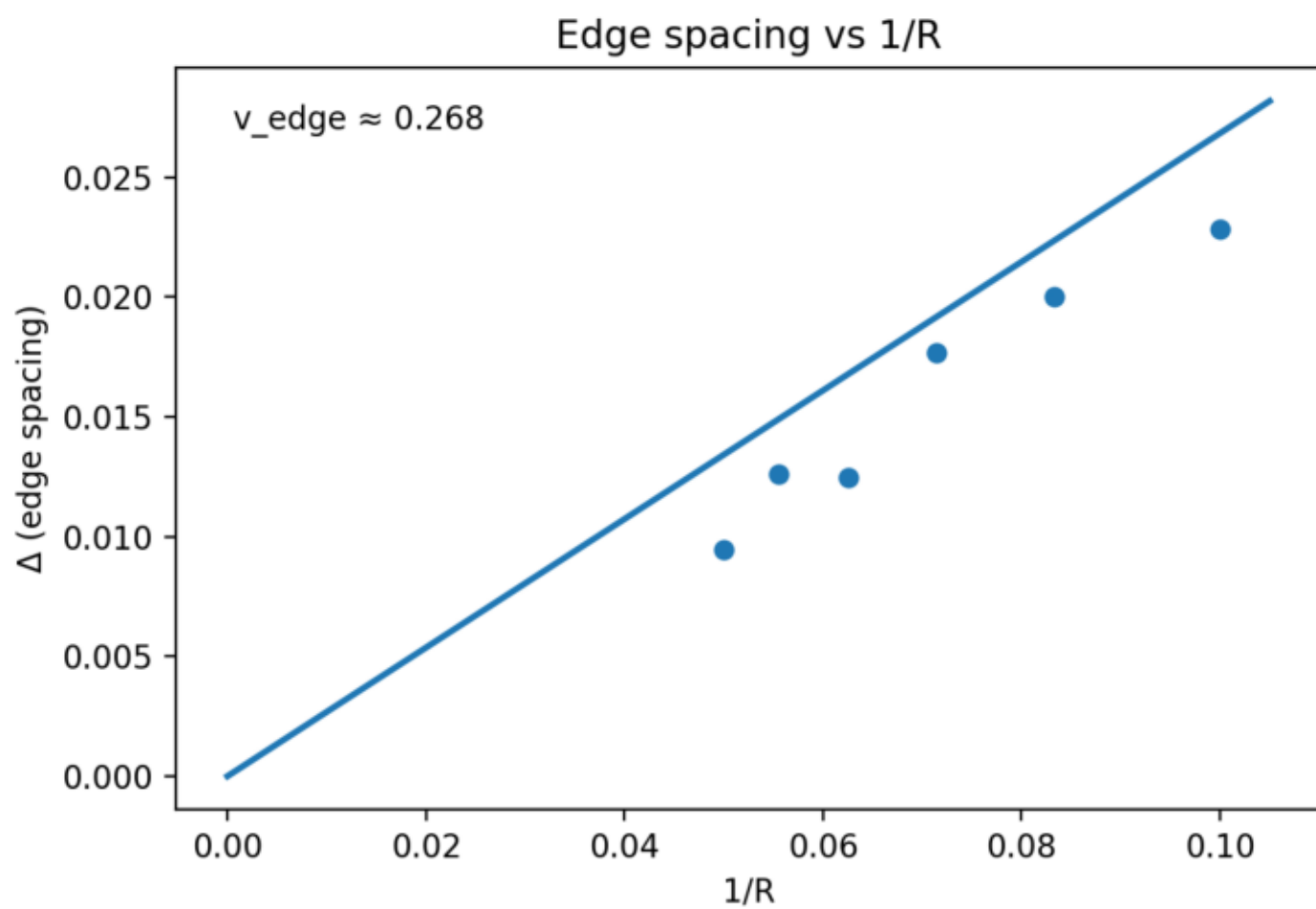


LDOS of representative edge state

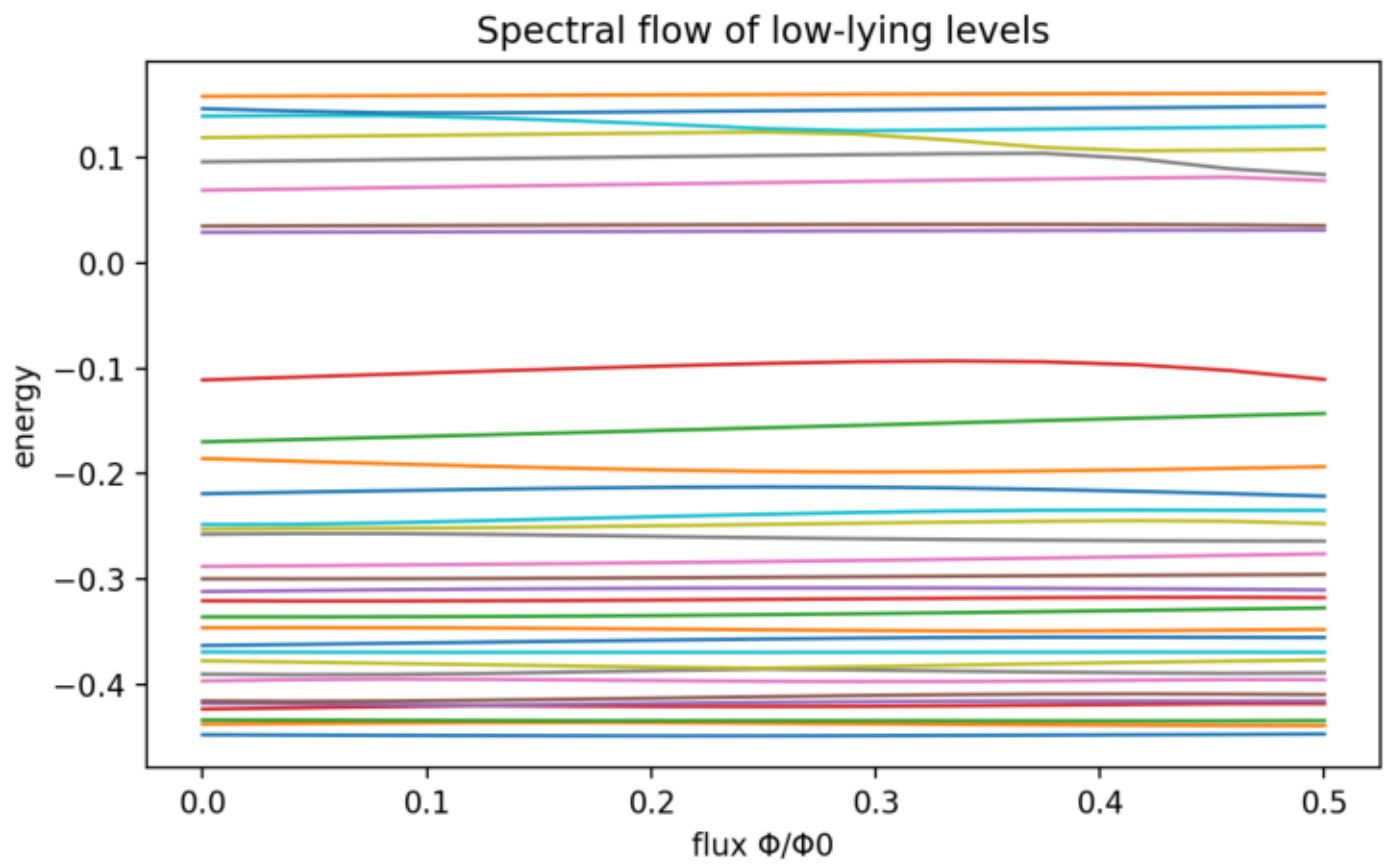
LDOS (edge state) @ $E=0.0293$



Edge spacing vs 1/R (fit)



Spectral flow under flux



Edge current map

